

SQL DEVELOPER INTERNSHIP

Task 1: Database Setup, Schema Design & Fundamentals

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Domain Selected: E-Commerce Management System
Tools Used: MySQL Workbench / SQLiteStudio

1. Schema Design SQL Script

The following DDL script defines the complete relational database structure for an E-Commerce domain, enforcing primary keys, foreign keys, auto-increment, unique constraints, and check constraints.

```
-- Create database structure
CREATE DATABASE IF NOT EXISTS ecommerce_db;
USE ecommerce_db;

-- 1. Users Table
CREATE TABLE users (
    user_id INT AUTO_INCREMENT PRIMARY KEY,
    full_name VARCHAR(100) NOT NULL,
    email VARCHAR(150) UNIQUE NOT NULL,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- 2. Categories Table
CREATE TABLE categories (
    category_id INT AUTO_INCREMENT PRIMARY KEY,
    category_name VARCHAR(100) NOT NULL UNIQUE
);

-- 3. Products Table
CREATE TABLE products (
    product_id INT AUTO_INCREMENT PRIMARY KEY,
    category_id INT NOT NULL,
    product_name VARCHAR(150) NOT NULL,
    price DECIMAL(10,2) CHECK (price >= 0),
    stock_quantity INT DEFAULT 0,
    FOREIGN KEY (category_id) REFERENCES categories(category_id) ON DELETE CASCADE
);

-- 4. Orders Table
CREATE TABLE orders (
    order_id INT AUTO_INCREMENT PRIMARY KEY,
    user_id INT NOT NULL,
    order_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    status VARCHAR(50) DEFAULT 'Pending',
    FOREIGN KEY (user_id) REFERENCES users(user_id) ON DELETE CASCADE
);

-- 5. Order_Items Table (Junction Table for M:N Relationship)
CREATE TABLE order_items (
    order_id INT NOT NULL,
    product_id INT NOT NULL,
```

```
quantity INT CHECK (quantity > 0),
unit_price DECIMAL(10,2) NOT NULL,
PRIMARY KEY (order_id, product_id), -- Composite Key
FOREIGN KEY (order_id) REFERENCES orders(order_id) ON DELETE CASCADE,
FOREIGN KEY (product_id) REFERENCES products(product_id) ON DELETE CASCADE
);
```

2. Conceptual Questions & Answers

Q1. What is normalization?

Normalization is the systematic process of organizing data in a relational database to reduce data redundancy, eliminate insert/update/delete anomalies, and ensure data integrity by splitting large tables into smaller, related entities.

Q2. Explain primary vs foreign key.

- **Primary Key:** A column (or combination of columns) that uniquely identifies each record in a table. It must be unique and cannot contain NULL values.
- **Foreign Key:** A column in one table that references the Primary Key of another table, establishing and enforcing a logical link between data across tables.

Q3. What are constraints?

Constraints are rules applied to table columns to enforce data validity, accuracy, and consistency. Common types include NOT NULL, UNIQUE, PRIMARY KEY, FOREIGN KEY, CHECK, and DEFAULT.

Q4. What is a surrogate key?

A surrogate key is an artificially generated unique identifier for a table row that has no business context or inherent meaning (e.g., an auto-incrementing integer like `user_id`).

Q5. How do you avoid data redundancy?

Data redundancy is minimized by applying database normalization principles (moving repeated data into separate lookup tables), using foreign key references, and avoiding duplicate attributes across tables.

Q6. What is an ER diagram?

An Entity-Relationship (ER) diagram is a visual model that illustrates the structure of a database by showing entities (tables), attributes (columns), and relationships (links) between entities.

Q7. What are the types of relationships in DBMS?

- **One-to-One (1:1):** A single row in Table A relates to at most one row in Table B.
- **One-to-Many (1:M):** A single row in Table A relates to multiple rows in Table B.
- **Many-to-Many (M:N):** Multiple rows in Table A relate to multiple rows in Table B (implemented via a junction table).

Q8. Explain the purpose of AUTO_INCREMENT.

AUTO_INCREMENT automatically generates a unique sequential integer whenever a new row is inserted, simplifying key assignment and ensuring unique identification.

Q9. What is the default storage engine in MySQL?

InnoDB is the default storage engine in MySQL. It provides ACID-compliant transaction support, foreign key constraints, row-level locking, and crash recovery.

Q10. What is a composite key?

A composite key is a primary key that consists of two or more columns combined together to uniquely identify a record, such as (`order_id`, `product_id`) in order item junction tables.